[Def.
Let V be a real or complex vector space.
A function $f: V \times V \to F$ is called a sesqui-linear form if it satisfies that
for all $\alpha, \beta, \gamma \in V$ and $c \in F$, $\begin{cases} f(c\alpha+\beta, \gamma) = c \cdot f(\alpha, \gamma) + f(\beta, \gamma) \\ f(\alpha, c\beta+\gamma) = \overline{c} \cdot f(\alpha, \beta) + f(\alpha, \gamma) \end{cases}$
In this page, we call the f as form, for

[Thm.
Let V be a finite dimensional inner product space and $f: V \times V \to F$ be a form.
Then, there is a unique linear operator T on V such that

$$f(\alpha, \beta) = \langle T\alpha, \beta \rangle \text{ for all } \alpha, \beta \in V.$$

Moreover, the map $f \mapsto T$ is isomorphism from space of forms onto $\mathcal{L}(V, V)$.
Proof Let $\beta \in V$ be fixed. Then, a map $\alpha \mapsto f(\alpha, \beta)$ is a linear map on V.
By the Riesz-Representation theorem, for an orthonormal basis $\beta = \{\alpha_1, \cdots, \alpha_n\}$,
there exists a unique vector $\beta' = \overline{f(\alpha_1, \beta)} \cdot \alpha_1 + \cdots + \overline{f(\alpha_n, \beta)} \cdot \alpha_n$ such that

$$\langle \alpha, \beta' \rangle = f(\alpha, \beta) \text{ for all } \alpha \in V.$$

Define $U: V \to V; \beta \mapsto \beta'$. Then, given $\alpha, \beta, \gamma \in V$ and $c \in F$,

$$\langle \alpha, U(c\cdot\beta+\gamma) \rangle = f(\alpha, c\beta+\gamma) = \overline{c} \cdot f(\alpha, \beta) + f(\alpha, \gamma) = \overline{c} \cdot \langle \alpha, U(\beta) \rangle + \langle \alpha, U(\gamma) \rangle = \langle \alpha, cU(\beta) + U(\gamma) \rangle$$

Since the α was arbitrary, it proves U is linear.
Moreover, since V is finite-dimensional, so the adjoint exists, put $T = U^*$.
Consequently, $f(\alpha, \beta) = \langle \alpha, U(\beta) \rangle = \langle T\alpha, \beta \rangle$.] Existence.

Corollary. Given forms $f, g : V \times V \to F$, define

$$\langle f, g \rangle \overset{\text{def}}{=} \operatorname{Tr}\left(T_f T_g^*\right) \overset{(\star)}{=} \sum_{j,k} f(\alpha_k, \alpha_j) \cdot \overline{g(\alpha_k, \alpha_j)}$$

is an inner product on $V \times V$ such that $(\star)$, for any orthonormal basis $\{\alpha_1, \dots, \alpha_n\}$ for V.

Proof. Given orthonormal basis $\beta = \{\alpha_1, \dots, \alpha_n\}$, let $A = [T_f]_\beta$ and $B = [T_g]_\beta$.

Then, by the orthonormality and the above Thm,

$$A_{jk} = \langle T_f(\alpha_k), \alpha_j \rangle = f(\alpha_k, \alpha_j) \quad \text{and} \quad B_{jk} = \langle T_g(\alpha_k), \alpha_j \rangle = g(\alpha_k, \alpha_j).$$

Moreover, $[T_f \circ T_g^*]_\beta = [T_f]_\beta \cdot [T_g^*]_\beta = AB^*$, so $\operatorname{Tr}(AB^*) = \sum_{j,k} A_{jk} \cdot \overline{B_{jk}}$ gives $(\star)$.

Def. Let V be a fin. dim inner product space, with basis $\beta = \{\alpha_1, \dots, \alpha_n\}$.

For a given form $f : V \times V \to F$, the matrix with entries

$$A_{jk} = f(\alpha_k, \alpha_j)$$

is called the matrix of f in the ordered basis β.

Observe that: for a form $f : V \times V \to F$ and orthonormal basis $\beta = \{\alpha_1, \dots, \alpha_n\}$.

Then, $f\left(\sum_s x_s \cdot \alpha_s, \sum_r y_r \alpha_r\right) = \sum_{r,s} \overline{y_r} A_{rs} x_s$, or

$$f(\alpha, \beta) = Y^* A X \quad \text{where } X = [\alpha]_\beta, \ Y = [\beta]_\beta.$$

Thm. Let V be a fin. dim Complex inn. Prod space.

For any form on $V \times V$, there is an orthonormal basis for V,

in which the matrix of f is upper-triangular.

Proof. Since V is fin. dim. Complex. inn. Prod. space, for linear operator $T : V \to V$ s.t.

$$f(\alpha, \beta) = \langle T(\alpha), \beta \rangle,$$

it has an orthonormal basis $\beta = \{\alpha_1, \dots, \alpha_n\}$ for V s.t. $[T]_\beta$ is upper-triangular. So,

$$f(\alpha_k, \alpha_j) = \langle T(\alpha_k), \alpha_j \rangle = 0 \quad \text{if } j > k.$$

Def. A form f on V is called Hermitian if

$$f(\alpha,\beta)=\overline{f(\beta,\alpha)} \quad \text{for all } \alpha,\beta\in V.$$

Note: if there is a linear operator T on V such that $f(\alpha,\beta)=\langle T(\alpha),\beta\rangle$, then $\overline{f(\beta,\alpha)}=\langle \alpha, T(\beta)\rangle=\langle T^{*}(\alpha),\beta\rangle$, so,

f is Hermitian if and only if T is Hermitian.

If T is Hermitian, then $\langle T(\alpha),\alpha\rangle$ is real for all $\alpha\in V$, so $f(\alpha,\alpha)$ is real.

Lemma. Let V be a complex inner product space, and f is a form on V such that

$$f(\alpha,\alpha) \text{ is real, for all } \alpha\in V.$$

Then, f is Hermitian.

Proof. Given $\alpha,\beta\in V$,

$$f(\alpha+\beta,\alpha+\beta)=f(\alpha,\alpha)+f(\beta,\beta)+f(\alpha,\beta)+f(\beta,\alpha)$$
$$f(\alpha+i\beta,\alpha+i\beta)=f(\alpha,\alpha)+f(i\beta,i\beta)-i\cdot f(\alpha,\beta)+if(\beta,\alpha)$$

By the assumption, $f(\alpha,\beta)+f(\beta,\alpha)$ and $-i\cdot f(\alpha,\beta)+i\cdot f(\beta,\alpha)$ are real. Thus,

$$f(\alpha,\beta)+f(\beta,\alpha)=\overline{f(\alpha,\beta)}+\overline{f(\beta,\alpha)} \quad \text{and} \quad -i\cdot f(\alpha,\beta)+i\cdot f(\beta,\alpha)=i\cdot\overline{f(\alpha,\beta)}-i\cdot\overline{f(\alpha,\beta)}$$

Multiplying i to the right, then we obtain

$$f(\alpha,\beta)-f(\beta,\alpha)=-\overline{f(\alpha,\beta)}+\overline{f(\alpha,\beta)}$$

So, $2f(\alpha,\beta)=2\cdot\overline{f(\alpha,\beta)}$, or f is Hermitian. ∎

Combining the above note with this Lemma, if V is fin. dim, then

T is Hermitian if and only if $\langle T(\alpha),\alpha\rangle$ is real.

[Principal Axis Theorem.]
[Thm. For every Hermitian form f on a fin. dim inn. prod space V,
there exists an orthonormal basis $\beta = \{\alpha_1, \cdots, \alpha_n\}$ such that

f is represented by a diagonal matrix with real entries.

Pf. Since V is fin. dim, there exists a linear operator T on V s.t

$$f(\alpha, \beta) = \langle T(\alpha), \beta \rangle.$$

And, by the lemma above, f is Hermitian iff T is Hermitian, so
there is an orthonormal basis for V which is consisted by eigenvectors of T.
Denote $\beta = \{\alpha_1, \cdots, \alpha_n\}$, and c_i be the eigenvalue corresponding to α_i. Then,

$$f(\alpha_k, \alpha_j) = \langle T(\alpha_k), \alpha_j \rangle = c_k \cdot \delta_{kj},$$

thus proved. (c_k are real by T is Hermitian). ☒
Note: under the condition hold,

$$f\left(\sum x_j \alpha_j, \sum y_k \alpha_k\right) = \sum_i c_i x_i \overline{y_i}$$

[HK. 9.2. (324P)

[Def.

A sesqui-linear form f is said to be non-degenerate if: for fixed $\alpha \in V$,

if $f(\alpha, \beta) = 0$ for all $\beta \in V$, then $\alpha = 0$.

[Prop.

Let $f : V \times V \to F$ be a form on an inner product space V.

f is non-degenerate if and only if the associated linear T_f is non-singular.

Proof. Suppose that f is non-degenerate. To show T_f is non-singular, ($\ker T_f = \{0\}$) assume that $T_f(\alpha) = 0$. That is, for any $\beta \in V$,

$$f(\alpha, \beta) = \langle T_f(\alpha), \beta \rangle = \langle 0, \beta \rangle = 0$$

So, $\alpha = 0$ by definition. Conversely, if T_f is non-singular, then for any non-zero $\alpha \in V$,

$$f(\alpha, \beta) = \langle T_f(\alpha), \beta \rangle \neq 0 \quad \text{if } \beta = T_f(\alpha)$$

which proves f is non-degenerate. □

Prop. (inn. prod ↓ space)

Let $f: V \times V \to F$ be a non-degenerate form on a finite dimensional space V. Given linear functional L, there exists one and only one vector $\beta \in V$ such that

$$L(\alpha) = f(\alpha, \beta) \text{ for all } \alpha \in V.$$

Proof By the theorem above, f has invertible associated linear map T such that

$$f(\alpha, \gamma) = \langle T(\alpha), \gamma \rangle \text{ for all } \alpha, \gamma \in V.$$

Meanwhile, by the Riesz Representation theorem, for the given L, there exists a unique vector $\beta' = \overline{L(\alpha_1)} \cdot \alpha_1 + \cdots + \overline{L(\alpha_n)} \alpha_n$, ($\{\alpha_1, \cdots, \alpha_n\}$ is orthonormal)

$$L(\alpha) = \langle \alpha, \beta' \rangle \text{ for all } \alpha \in V.$$

Now, let $\beta = (T^*)^{-1}(\beta')$, then

$$f(\alpha, \beta) = \langle T(\alpha), \beta \rangle = \langle T(\alpha), (T^*)^{-1}(\beta') \rangle = \langle \alpha, (T^*) \circ (T^*)^{-1}(\beta') \rangle = \langle \alpha, \beta' \rangle = L(\alpha).$$

And, the uniqueness is given by: $f(\alpha, \beta_1) = L(\alpha) = f(\alpha, \beta_2) \Rightarrow f(\alpha, \beta_1 - \beta_2)$ and since f is non-degenerate, so $\beta_1 - \beta_2$ must be zero. (Let $\alpha = \beta_1 - \beta_2$) ∎

Prop.

Let $f: V \times V \to F$ be a non-degenerate form on a finite dimensional inner Product space V. Given a linear operator $S: V \to V$, there exists an adjoint S' relative to f. (That is, $f(S\alpha, \beta) = f(\alpha, S'\beta)$ for all $\alpha, \beta \in V$).

Proof